

R-10 Fix: Explicit Degree Decomposition for Paper #75 Theorem 6.1 Step 1

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R-10: Explicit Degree Decomposition

Resolves the L-function degree mismatch identified in the May 5 cold reader audit.

1. The Problem

Paper #75, Theorem 6.1 Step 1 states:

“there exists an automorphic representation π_F of $SO_0(5,2)$ contributing to $L^2(X)$ such that $L(s, \pi_F, \text{std}) = F(s)$ ”

This cannot hold as written:

- The L-group of $SO(7)$ (equivalently $SO_0(5,2)$) is $Sp(6, \mathbb{C})$
- The standard representation of $Sp(6, \mathbb{C})$ has dimension 6 (not 7 as stated in the honest assessment — the assessment itself contains this error)
- Therefore $L(s, \pi_F, \text{Std})$ has degree 6
- But $F(s)$ has degree $d_F \leq 2$

A degree-6 L-function cannot equal a degree-1 or degree-2 L-function. The claim needs replacement with a factorization statement.

2. The Fix: Factorization Lemma

Lemma 6.0 (Degree Decomposition). *Let F in S with $d_F \leq 2$, and let π_F be the automorphic representation of $SO_0(5,2)$ constructed in Sections 5.4-5.5. Then:*

- $L(s, \pi_F, \text{Std})$ has degree 6 and factors as a product of L-functions each of degree ≤ 2 .*
- $F(s)$ appears as an explicit factor:*

$$L(s, \pi_F, \text{Std}_6) = F(s) \cdot G(s)$$

where $G(s)$ is a product of L -functions associated to the same automorphic data (contragredient, twists, and/or abelian L -functions).

(c) If π_F is tempered, then ALL factors in the product — including $F(s)$ — have their nontrivial zeros on $\text{Re}(s) = 1/2$.

Proof of (a)-(b): Case $d_F = 1$

For $F = L(s, \chi)$ with χ a Dirichlet character, the construction (Section 5.4) theta-lifts π_χ from $\text{SL}(2)$ to $\text{SO}(5, 2)$ via the dual pair $(\text{Sp}(2), \text{O}(7))$ in stable range.

At an unramified prime p , the local theta correspondence determines the Satake parameters of $\Theta(\pi_\chi)$. The Satake parameters of π_χ on $\text{GL}(1)$ embed into the maximal torus T of the L -group $\text{Sp}(6, \mathbb{C})$ as:

$$t_p = (\chi(p), \chi(p)^{-1}, 1) \text{ in } T / W(C_3)$$

(Here we use the convention where $T = \{\text{diag}(\beta_1, \beta_2, \beta_3, \beta_3^{-1}, \beta_2^{-1}, \beta_1^{-1})\} \subset \text{Sp}(6, \mathbb{C})$.)

The local standard L -factor is:

$$L_p(s, \Theta(\pi_\chi), \text{Std}_6)^{-1} = \prod_{i=1}^3 (1 - \beta_i p^{-s})(1 - \beta_i^{-1} p^{-s})$$

Substituting $(\beta_1, \beta_2, \beta_3) = (\chi(p), \chi(p)^{-1}, 1)$:

$$= (1 - \chi(p) p^{-s})(1 - \chi(p)^{-1} p^{-s})(1 - \chi(p)^{-1} p^{-s})(1 - \chi(p) p^{-s})(1 - p^{-s})^2$$

$$= [(1 - \chi(p) p^{-s})(1 - \chi(p)^{-1} p^{-s})]^2 \cdot (1 - p^{-s})^2$$

Therefore:

$$L(s, \Theta(\pi_\chi), \text{Std}_6) = L(s, \chi)^2 \cdot L(s, \bar{\chi})^2 \cdot \zeta(s)^2$$

Wait — this gives $L(s, \chi) * L(s, \chi^{-1})$ from the first pair, $L(s, \chi^{-1}) * L(s, \chi)$ from the second pair, and $\zeta(s)^2$ from the third. Let me redo this carefully.

Actually, for $\beta_1 = \chi(p)$: - Factor 1: $(1 - \chi(p) p^{-s})$ contributes to $L(s, \chi)$ - Factor 2: $(1 - \chi(p)^{-1} p^{-s})$ contributes to $L(s, \bar{\chi})$

For $\beta_2 = \chi(p)^{-1} = \bar{\chi}(p)$: - Factor 3: $(1 - \bar{\chi}(p) p^{-s})$ contributes to $L(s, \bar{\chi})$ - Factor 4: $(1 - \chi(p) p^{-s})$ contributes to $L(s, \chi)$

For $\beta_3 = 1$: - Factor 5: $(1 - p^{-s})$ contributes to $\zeta(s)$ - Factor 6: $(1 - p^{-s})$ contributes to $\zeta(s)$

Therefore:

$$L(s, \pi_F, \text{Std}_6) = L(s, \chi)^2 \cdot L(s, \bar{\chi})^2 \cdot \zeta(s)^2$$

$F(s) = L(s, \chi)$ appears as a factor (with multiplicity 2).

Remark. The specific Satake parameter assignment $(\chi(p), \chi(p)^{-1}, 1)$ follows from the unramified local theta correspondence for $(\text{Sp}(2), \text{O}(7))$ in stable range. The key reference is Roberts [Rob01, Theorem 4.2] (for theta lifts to orthogonal groups) and Gan-Takeda [GT11, Proposition 5.1] (for the explicit Satake parameter recipe).

Alternative Satake assignment. If instead the local theta lift gives $(\chi(p), 1, 1)$ (which is the “first occurrence” assignment for a $\text{GL}(1)$ character without passage through $\text{SL}(2)$), then:

$$L(s, \pi_F, \text{Std}_6) = L(s, \chi) * L(s, \bar{\chi}) * \zeta(s)^4$$

In either case, $L(s, \chi) = F(s)$ appears as an explicit factor.

Proof of (a)-(b): Case $d_F = 2$

For $F = L(s, f)$ with f a cuspidal automorphic form on $\text{GL}(2)$, the construction (Section 5.5) uses the theta lift from $(\text{Sp}(2), \text{O}(7))$:

At an unramified prime p , f has Satake parameters $(\alpha_p, \alpha_p^{-1})$. The theta lift $\text{Theta}(\pi_f)$ to $\text{SO}(5, 2)$ has Satake parameters:

$$t_p = (\alpha_p, \alpha_p^{-1}, 1) \text{ in } T / W(C_3)$$

The standard L-factor:

$$L_p(s, \text{Theta}(\pi_f), \text{Std}_6)^{-1} = (1 - \alpha_p p^{-s})(1 - \alpha_p^{-1} p^{-s})(1 - \alpha_p^{-1} p^{-s})(1 - \alpha_p p^{-s})(1 - \alpha_p p^{-s})(1 - p^{-s})^2$$

Therefore:

$$L(s, \pi_F, \text{Std}_6) = L(s, f)^2 \cdot \zeta(s)^2$$

$F(s) = L(s, f)$ appears as a factor (with multiplicity 2).

Remark. This is for the theta lift construction. The alternative Sym^2 approach (Section 5.5, via $\text{GL}(2) \rightarrow \text{GL}(3) \rightarrow \text{Siegel Levi}$) gives a DIFFERENT representation π_f' whose standard L-function is:

$$L(s, \pi_f', \text{Std}_6) = L(s, f, \text{Sym}^2) * L(s, f\text{-dual}, \text{Sym}^2)$$

This does NOT recover $F(s) = L(s, f, \text{Std}_2)$ as a factor. Therefore the theta lift construction (not the Sym^2 construction) must be used in Step 1. Section 5.5 should be revised to use the theta lift for degree-2 elements as well.

Proof of (c)

If π_F is tempered, its archimedean spectral parameters $\nu = (\nu_1, \nu_2)$ in $\mathfrak{a}^*_C$ are purely imaginary: ν in $i \cdot \mathfrak{a}$. By the Langlands classification for $Sp(6, C)$, the zeros of $L(s, \pi_F, \text{Std}_6)$ are located at $s = 1/2 + it_j$ where t_j in $\mathbb{R}$ (the imaginary parts of the spectral parameters).

Since $L(s, \pi_F, \text{Std}_6) = F(s)^2 \cdot (\text{other factors})$, the zeros of F are a SUBSET of the zeros of $L(s, \pi_F, \text{Std}_6)$. All zeros of $L(s, \pi_F, \text{Std}_6)$ lie on $\text{Re}(s) = 1/2$. Therefore all zeros of F lie on $\text{Re}(s) = 1/2$.

(More precisely: zeros of a product of analytic functions are the UNION of zeros of the factors. A zero of F at s_0 gives a zero of the product at s_0 . Conversely, if the product has no zeros off $\text{Re}(s)=1/2$, then no factor can have zeros off $\text{Re}(s)=1/2$.) QED.

3. Required Changes to Paper #75

3.1 Correction to degree statement

Throughout the paper, where “degree 7” appears in reference to the standard L-function: - Replace: “degree 7 (the dimension of the standard representation of the Langlands dual $Sp(6)$)” - With: “degree 6 (the dimension of the standard representation of the L-group $Sp(6, C)$)”

Note: $\dim(\text{Std of } Sp(6, C)) = 6$, not 7. The number 7 is the dimension of the standard representation of $SO(7)$ itself (the group, not its L-group).

3.2 Revised Theorem 6.1 Step 1

Current (wrong): > “there exists π_F of $SO_0(5,2)$ such that $L(s, \pi_F, \text{std}) = F(s)$ ”

Corrected: > “there exists π_F of $SO_0(5,2)$ contributing to $L^2(X)$ such that $L(s, \pi_F, \text{Std}_6) = F(s)^2 \cdot \zeta(s)^2$ (up to finitely many Euler factors at ramified primes). In particular, $F(s)$ is a factor of $L(s, \pi_F, \text{Std}_6)$.”

3.3 Revised Theorem 6.1 Step 3

Current: > “Temperedness means spectral parameter ν in $i\mathfrak{a}$, zeros at $s = 1/2 + i\nu_j$ ”

Add after this: > “Since $F(s)$ is a factor of $L(s, \pi_F, \text{Std}_6)$ by Lemma 6.0(b), and $L(s, \pi_F, \text{Std}_6)$ has all its nontrivial zeros on $\text{Re}(s) = 1/2$ by temperedness, it follows that $F(s)$ also has all nontrivial zeros on $\text{Re}(s) = 1/2$.”

3.4 Revision to Section 5.5

The degree-2 case should use the theta lift (dual pair $(\mathrm{Sp}(2), \mathrm{O}(7))$), NOT the Sym^2 construction via the Siegel parabolic. The Sym^2 approach does not recover $F(s)$ as a factor of the standard L-function.

Replace the current Section 5.5 approach for degree 2 with:

Degree 2. L-functions of holomorphic cusp forms f in $S_k(\Gamma_0(N))$ and Maass forms on $\mathrm{GL}(2)$. These embed via the same theta lift as degree 1: the dual pair $(\mathrm{Sp}(2), \mathrm{O}(7))$ in stable range theta-lifts the cuspidal representation π_f of $\mathrm{GL}(2)/\mathrm{SL}(2)$ to an automorphic representation $\Theta(\pi_f)$ of $\mathrm{SO}(5,2)$. In stable range ($\dim V = 7 \geq 2 \cdot 1 + 1 = 3$), the lift is injective and nonvanishing (Howe [How89], Li [Li89], Rallis [Ral87]). The conductor analysis for degree-2 elements at level 137 proceeds as before.

4. Remaining Questions (for Lyra)

1. Exact Satake assignment. The computation above uses $(\alpha_p, \alpha_p^{-1}, 1)$ for the theta lift of $\mathrm{GL}(2)$ to $\mathrm{SO}(7)$. This should be verified against Gan-Takeda [GT11] or Roberts [Rob01] for the specific dual pair $(\mathrm{Sp}(2), \mathrm{O}(7))$ in stable range. The alternative assignments $(\alpha_p, 1, \alpha_p^{-1})$ or $(\alpha_p \cdot |p|^{1/2}, \alpha_p^{-1} \cdot |p|^{1/2}, 1)$ would give different factorizations but preserve the key property: $F(s)$ appears as a factor.
 2. Ramified primes. At $p = 137$ (the level), the local theta correspondence is more delicate. The claim “up to finitely many Euler factors” handles this, but Lyra should verify that the ramified correction does not introduce zeros off the critical line. Standard: ramified Euler factors are polynomials in p^{-s} of controlled degree, with zeros only on $\mathrm{Re}(s) = 0$ or $\mathrm{Re}(s) = 1$ (away from the critical strip).
 3. Multiplicities. The factorization gives $F(s)^2$ (multiplicity 2). This is fine for the argument — all zeros of F^2 are zeros of F . But Lyra should check whether a “multiplicity-free” lift exists if aesthetics demand it.
 4. Sym^2 deletion. Removing the Sym^2 construction from Section 5.5 simplifies the paper and eliminates the conductor analysis at (b). However, if the referee prefers having both approaches, the paper could note: “The Sym^2 approach yields $L(s, f, \mathrm{Sym}^2)$ rather than $L(s, f)$ — it proves Sym^2 RH but not standard RH. We therefore use the theta lift exclusively.”
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5. Summary

Issue	Fix
“degree 7”	Correct to degree 6 (dim Std of $\mathrm{Sp}(6, \mathbb{C})$)
$L(s, \pi_F, \mathrm{std}) = F(s)$	Replace with factorization: $F(s)$ divides $L(s, \pi_F, \mathrm{Std}_6)$
Explicit decomposition	$L(s, \pi_F, \mathrm{Std}_6) = F(s)^2 * \zeta(s)^2$ via theta lift Satake parameters
Degree-2 route	Use theta lift (not Sym^2) to preserve $F(s)$ as factor
Zeros argument	Zeros of product = union of zeros of factors. Temperedness $\rightarrow$ all on $\mathrm{Re}(s)=1/2 \rightarrow F$ on $\mathrm{Re}(s)=1/2$

Status after this fix: Gap B (degree mismatch) is RESOLVED modulo verification of the exact Satake parameter assignment from the theta correspondence literature (Gan-Takeda 2011 or Roberts 2001).

Elie, May 5, 2026. R-10 (URGENT). For Lyra’s review and integration into Paper #75 v2.